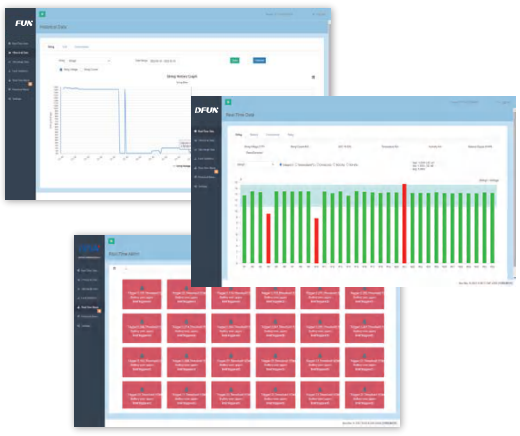
PBMS9000
for Data Center



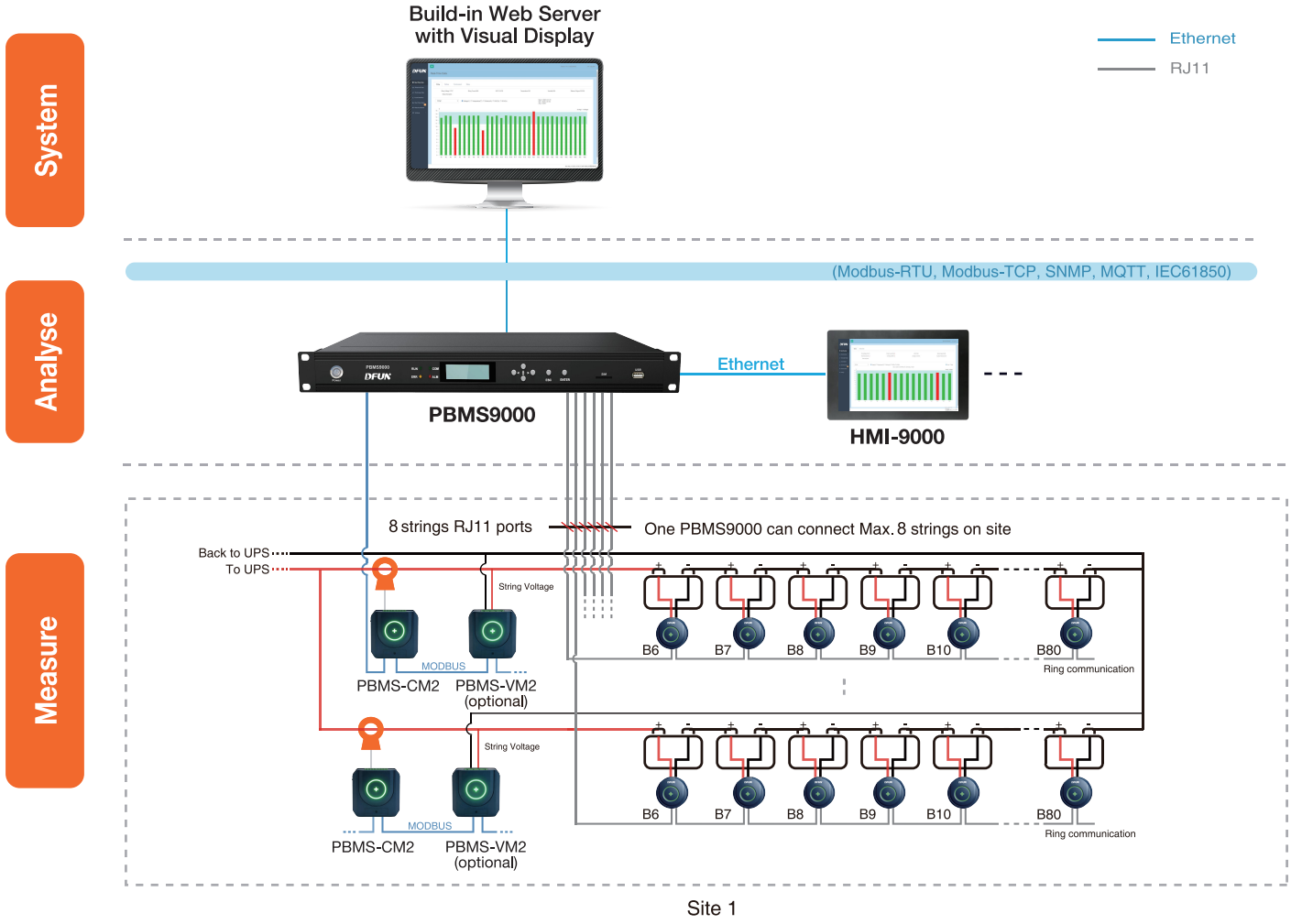
Feature

- Apply to UPS and data center application
- Measure lead-acid or multi-pole battery
- Ring communication, any communication failure will not affect other sensors communication
- Monitor battery voltage, current, impedance, insulation resistance, ripple current & voltage, SOC, SOH, etc.
- Support Modbus, SNMP, MQTT and IEC61850 protocols
- Auto-sensing for the battery sensor's ID address
- Dual-source to avoid power shutdown
- Anti-interference design, support to connect with high-frequency UPS
- Comply with IEEE 1188-2005
- Thermal runaway management(Optional temperature and humidity sensor)
- The product has obtained CE, FCC, Rohs, UL certification



PBMS9000 for Data Center

System Structure



Site 1

Management Layer

PBMS9000 Battery Management Controller

- Standard 1U design for distribution cabinets
- Dual-source to avoid power shutdown
- Built-in web server with visual display
- Monitors Max. 8 strings battery, in a total of 640 batteries
- Measures battery string current & voltage, ripple voltage & current, charge & discharge current, internal temperature, impedance, insulation resistance, ambient temperature & humidity, SOC and SOH
- Alarm notifications via SMS or Email
- Supports Modbus-TCP, Modbus-RTU, SNMP and IEC61850 protocols
- Supports MQTT for JSON format to data upload
- 1 RS485 port, 2 Ethernet ports and 1 4G antenna port for data upload
- 6 DI ports (for digital Input connecting)
- 6 DO ports (for sound and light alarms)

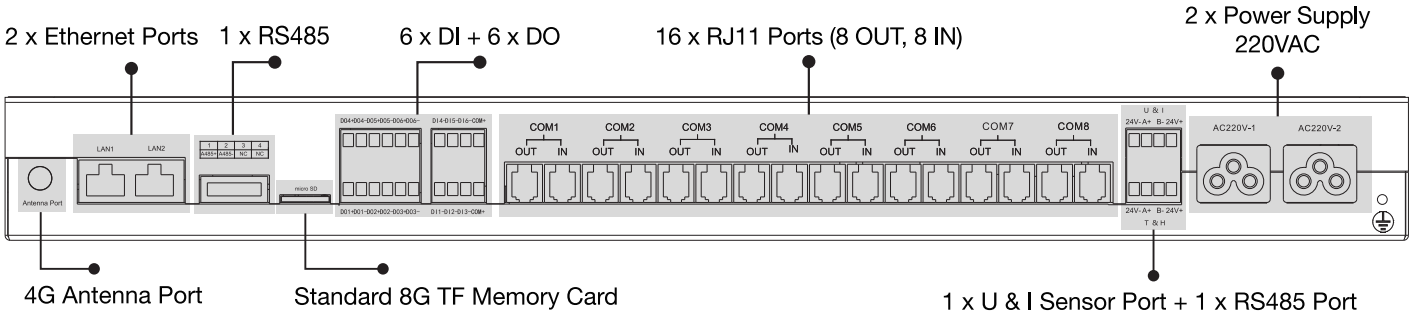


Pre-install & Pre-wiring for PBMS9000 and HMI-9000 (Optional)



PBMS9000 for Data Center

Interface Description



Technical Specification

CPU	ARM cortex-A7 528MHz	**Up-link Communication**	2 Ethernet ports(10/100M), MODBUS-TCP, SNMP, IEC61850 1RS485 port, MODBUS-RTU, baudrate: 9600bps/19200bps/38400bps(optional)	
Memory	512MB flash 4G EMMC + 8G TF memory card			
Display	2-inch LCD with backlight	**Down-link Communication**	8 Channels RJ11 ports, each port Max. connect ≤ 80pcs batteries, total Max. 640pcs	
MTBF	≥100,000 hours	**Measure Range**	Voltage	1~8 strings, Range: 20~1000V DC (±0.1%), Resolution: 0.01V
Power Supply	V1: 48VDC, Range: 18~72VDC V2: 220VAC, Range: 90~264VAC/127~370VDC V3: 110VDC, Range: 43~160VDC		Ripple Voltage	1~8 strings, Range: 2~100V (peak), Resolution: 0.01V
Dimension	Standard 19-inch 1 U device 483mm × 206mm × 44.5mm (W*D*H) Open Hole: 440mm×46mm (L*H)		Current	1~8 strings, Range: -3000~+3000A DC (±1%fs, 15°C~35°C), Resolution: 0.01A
Power Consumption	≤15W (Only main controller)		Ripple Current	1~8 strings, 20% of Hall sensor rated current (peak), Resolution:0.01A
Additional Port	6 x DI dry contact 6 x DO relay output, 250VAC/5A or 30VDC/5A	**Environment**	Range: -10~+55℃, Resolution:0.1℃, Accuracy: ± 0.5 ℃	
		Extension Sensor (Optional)	-RS485 for Max.8 strings ambient temperature & humidity measuring -RS485 for 1~8 strings DC insulation resistance measuring	

String Measuring Sensor

PBMS-CM2 String Current Measuring Sensor & Hall Sensor

- 1 string needs 1 PBMS-CM2, each PBMS-CM2 with 2 hall sensor ports
- Measure battery string charge and discharge current, ripple current
- Hall sensor measuring range: -3000A~+3000A(single-pole) / -6000A~+6000A(dual-pole)
- Supports floating charge hall detection



Hall sensor



Item	Power Supply	Hall Interface	Environment
PBMS-CM2	24VDC (range: 9~32VDC) Power consumption: <0.5W	2 channels, DC ±4V	-10˚C ~ +55˚C , accuracy: ±1˚C

PBMS9000 for Data Center

Battery Cell Sensor

PBAT61-02/PBAT61-06/PBAT61-12 Battery Cell Sensor

- PBAT61-02 for 2V battery
- PBAT61-06 for 6V battery
- PBAT61-12 for 12V battery
- Monitor individual battery voltage, internal temperature (negative pole), Impedance(ohmic value)
- Auto-balancing



Running Status

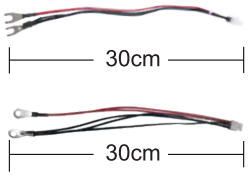


Abnormal Status

Technical Specification

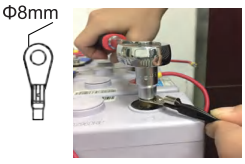
Item	Rated Input Voltage	Measuring Range			
		Voltage	Internal Temperature	Impedance	Power Consumption
PBAT61-02	02V	1.6~2.6V (±0.2%)	-20°C ~ 85°C (±0.5°C)	Range: 0.1~50mΩ Repeatability accuracy: 1.0%±25μΩ Consistency accuracy: 1.5%±25μΩ	Running: <70mW Sleeping: <8mW
PBAT61-06	06V	4.8~10V (±0.2%)			
PBAT61-12	12V	7.5~15.6V (±0.2%)			

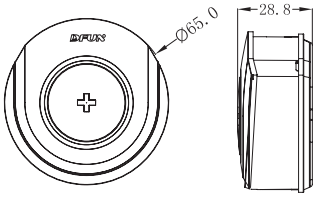
Dimension and Installation

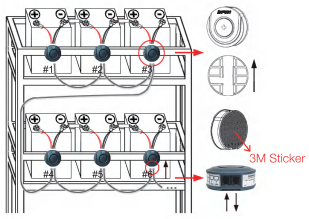


Battery Measuring Cable	Item	Description
	Length	30cm
	Terminal & Size	U type, hole diameter: 8mm
		O type, hole diameter: 8mm













Extension Module (Optional)

Ambient Temperature & Humidity Sensor



- Dual RJ45 interface, fast wiring, support cascade use
- Adsorption by magnet
- Temperature: -20°C ~ 60°C (±0.4°C), resolution: 0.1°C. Humidity: 0 ~ 100%RH (±3%RH), resolution: 0.1%RH
- Cable(9000-H-THD-RJ45-500): 5 meter RJ45 convert to 4P terminal wire

PBMS-VM2 String Voltage Measuring Sensor



- 1 PBMS-VM2 can measure 1 string's voltage
- Measure battery string voltage and ripple voltage
- Built-in ground fault detection & testing

10.1-inch Android HMI



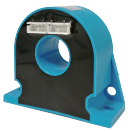
- 10.1" inch touch-screen HMI for local display and operation
- OS: Android 10.0
- CPU: R818 (4-core A53 1.6GHz)
- Running memory: 1GB,
- EMMC: 8GB
- Support the reading and configuration of all parameters of the system

Hydrogen (H2) Sensor



- Class of protection: IP65
- Measuring range: 0~1000ppm
- Accuracy: ±5%FS
- Resolution: 1ppm(100KΩ~50KΩ)
- Cable: 2 meter Aviation port to 4P terminal wire

DC Leakage Current Sensor



- Monitoring string leakage current to prevent fire caused
- Cable: 2 meter hall sensor port to 4P terminal wire
- Accuracy: ±1%FS

Order Information

<table>
<tr><th colspan="2">System Structure</th><th>Model</th><th colspan="3">Description</th><th>Remark</th></tr>
<tr><td rowspan="8">Management Layer</td><td rowspan="8">BMS
Main Controller</td><td colspan="4">PBMS9000-8C- ① - ② - ③ - ④</td><td rowspan="5">Monitors Max.8 strings (Max.640 batteries)</td></tr>
<tr><td rowspan="5">Mandatory</td><td rowspan="3">①
Power supply voltage</td><td>V1</td><td>48VDC</td></tr>
<tr><td>V2</td><td>220VAC</td></tr>
<tr><td>V3</td><td>110VDC</td></tr>
<tr><td rowspan="2">②
Power supply channel</td><td>1P</td><td>Single channel</td></tr>
<tr><td>2P</td><td>Two channels</td></tr>
<tr><td rowspan="3">Optional</td><td>③
Communication</td><td>L</td><td>Support 4G</td><td></td></tr>
<tr><td rowspan="2">④
Protocol
(Choose one option)</td><td>CP</td><td>Support IEC61850</td><td rowspan="2">Monitors Max.108 batteries</td></tr>
<tr><td>TC</td><td>Support IEC104</td></tr>
<tr><td colspan="2" rowspan="2">String Measuring Sensor</td><td>PBMS-CM2</td><td colspan="2">String current measuring sensor</td><td>One PBMS-CM2 per string, each is equipped with 2 Hall sensor ports, and can connect a maximum of 2 Hall sensors. If is used for multi-pole batteries, we can configure the PBMS-CM2 according to the number of poles
Hall sensor cable: 2m
Communication cable: 5m(with RJ45 port)</td></tr>
<tr><td>PBMS-VM2</td><td colspan="2">String voltage measuring sensor</td><td>One per string (optional)</td></tr>
<tr><td colspan="2" rowspan="3">Battery Cell Sensor</td><td>PBAT 61-02</td><td colspan="2">02V battery cell sensor</td><td>One per battery cell</td></tr>
<tr><td>PBAT 61-06</td><td colspan="2">06V battery cell sensor</td><td>One per battery cell</td></tr>
<tr><td>PBAT 61-12</td><td colspan="2">12V battery cell sensor</td><td>One per battery cell</td></tr>
<tr><td rowspan="15">Accessory</td><td rowspan="2">Cell Sensor Communication Cable</td><td>PBAT-COM-40</td><td colspan="3">Cable terminal: RJ11 port, length: 40cm (standard)</td></tr>
<tr><td>PBAT-COM-70</td><td colspan="3">Cable terminal: RJ11 port, length: 70cm</td></tr>
<tr><td rowspan="2">Battery Measuring Cable</td><td>PBAT-C1-U-M8-30</td><td colspan="3">Cable terminal: U type  , hole diameter: 8mm, length: 30cm (standard)</td></tr>
<tr><td>PBAT-C1-O-M8-30</td><td colspan="3">Cable terminal: O type  , hole diameter: 8mm, length: 30cm</td></tr>
<tr><td rowspan="11">Hall Sensor for PBMS-CM2</td><td>DXEEK03-50A-V4</td><td>Rated input: 50A</td><td>Measure Range: -50A~+50A, Φ20mm</td></tr>
<tr><td>DXEEK03-100A-V4</td><td>Rated input: 100A</td><td>Measure Range: -100A~+100A, Φ20mm</td></tr>
<tr><td>DXEEK05-25A-V4</td><td>Rated input: 25A</td><td>Measure Range: -25A~+25A, Φ40mm</td></tr>
<tr><td>DXEEK05-200A-V4</td><td>Rated input: 200A</td><td>Measure Range: -200A~+200A, Φ40mm</td></tr>
<tr><td>DXEEK05-300A-V4</td><td>Rated input: 300A</td><td>Measure Range: -300A~+300A, Φ40mm</td></tr>
<tr><td>DXEEK05-400A-V4</td><td>Rated input: 400A</td><td>Measure Range: -400A~+400A, Φ40mm</td></tr>
<tr><td>DXEEK05-500A-V4</td><td>Rated input: 500A</td><td>Measure Range: -500A~+500A, Φ40mm</td></tr>
<tr><td>DXEEK05-600A-V4</td><td>Rated input: 600A</td><td>Measure Range: -600A~+600A, Φ40mm</td></tr>
<tr><td>DXEEK05-800A-V4</td><td>Rated input: 800A</td><td>Measure Range: -800A~+800A, Φ40mm</td></tr>
<tr><td>DXEEK05-1000A-V4</td><td>Rated input: 1000A</td><td>Measure Range: -1000A~+1000A, Φ40mm</td></tr>
<tr><td rowspan="8">Optional Module</td><td rowspan="2">HMI for Local Display & Operation</td><td>HMI-9000</td><td>10-inch touch-screen HMI for local display & operation</td><td>One per six (max) PBMS9000
Communication with PBMS9000 via Ethernet</td></tr>
<tr><td>HDR-30-12</td><td>12VDC power coverter (220VAC to 12VDC)</td><td>If selected HMI-PBMS9000, then must select power supply converter</td></tr>
<tr><td>DC Leakage Current Sensor</td><td>STL8S-MT2-300mA-DF</td><td>Rated Input: ±300mA</td><td>Measure Range:-360mA~+360mA, Φ72mm</td></tr>
<tr><td>Ambient Temperature & Humidity</td><td>H-THDRJ45</td><td>Temperature: -20°C ~ 60°C (±0.4°C)
Humidity: 0 ~ 100%RH (±3%RH)</td><td>One PBMS9000 can conncet Max. 6pcs</td></tr>
<tr><td>Hydrogen (H_2) Sensor</td><td>PBMS-H_2</td><td>Measuring Range: 0~1000ppm</td><td>One per one PBMS9000</td></tr>
<tr><td rowspan="2">Cabinet for Prewiring In Factory</td><td>PBMS-BOX-260</td><td>Dimension: 535*260*335mm (W*H*D)</td><td>Pre-install & pre-wiring for PBMS9000 and HMI-9000</td></tr>
<tr><td>PBMS-BOX-730</td><td>Dimension: 535*730*335mm (W*H*D)</td><td>Pre-install & pre-wiring for Max.4 PBMS9000 and one HMI-9000</td></tr>
<tr><td>Ethernet Switch</td><td>IES318</td><td>1. Power supply 24VDC 12~48VDC
2. 8 x 10/100M Ethernet ports</td><td>If selected PBAT-BOX-730, then must select IES318</td></tr>
</table>

Note:

① If battery terminal is M5, M6, M10, M12, M16, please contact our sales for customize requirements before placing the order.

② If the Hall sensor measuring range exceeds -1000~+1000A, please contact our sales for customization before ordering.

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Your Representative